An Alternating Guidance With Cross-View Teacher–Student Framework for Remote Sensing Semi-Supervised Semantic Segmentation

Yujia Fu, Mingyang Wang, Gemine Vivone, *Senior Member, IEEE*, Yunhong Ding, and Lin Zhang

***Abstract*— The semantic segmentation of remote sensing images is crucial for Earth observation. The semi-supervised semantic segmentation method can effectively reduce the dependence of the training process on labeled data. Among them, the semi-supervised semantic segmentation method based on the teacher–student paradigm is currently one of the most mainstream methods. However, the issue of weight coupling has constrained further performance improvements. This article proposes an alternating guidance method that combines cross-view learning to improve the teacher–student paradigm and enhance the semantic segmentation performance of remote sensing images. The student model is designed by using two decoders with the same architecture but independently updated parameters. Two decoders process the input obtained after image and feature level perturbations. This allows the student model to generate unique feature representations and enhances its learning capability. The teacher model uses two decoders to construct an alternating supervision mechanism. The two decoders of the teacher model take turns outputting pseudo-labels to guide the training process of the student model. This alternating supervision strategy can provide richer supervision signals for student model while helping to alleviate weight coupling between teacher and student. The experiments on two remote sensing image datasets show that compared with the state-of-the-art (SOTA) semi-supervised semantic segmentation methods, the method proposed demonstrates excellent competitiveness.**

***Index Terms*— Alternating guidance, cross-view learning, deep learning, image segmentation, remote sensing, semantic segmentation, semi-supervised learning.**

I. Introduction

SEMANTIC segmentation of remote sensing images is a pivotal task within the domain of remote sensing image processing, aiming to assign specific semantic labels to each pixel in an image. This technique has been widely applied across numerous research fields [1], [2], [3], including land cover classification [4], [5], [6], [7], urban planning [8], [9], and disaster monitoring [10]. In recent years, deep learning methodologies have demonstrated remarkable performance in semantic segmentation tasks. However, fully supervised learning paradigms are data-hungry, requiring extensively annotated datasets. For large-scale remote sensing datasets, the annotation process is both time-consuming and labor-intensive. To address this issue, semi-supervised learning approaches have recently gained prominence. This method leverages a small amount of labeled data together with a large amount of unlabeled data, thus alleviating the limitation associated with data annotation.

The principle challenge in semi-supervised semantic segmentation lies in the effective exploitation of vast quantities of unlabeled data to extract useful information that improves the training effect of the model. Pseudo-label [11] techniques address it by generating pseudo-labels for unannotated images and gradually incorporating these labels into the training process, thereby providing supervision to the model. On other hand, consistency regularization methods [12], [13], [14], [15] learn unlabeled data by enforcing the model to make consistent predictions on data after different perturbations, encompassing data augmentation, feature perturbation, and network perturbation. Recent state-of-the-art (SOTA) approaches leverage the teacher–student paradigm [16], [17], [18], [19], [20], [21]. In this method, both the teacher and student models share identical architectures, with the teacher's predictions serving as guidance for the student's training. The parameters of the student model are updated through backpropagation, while those of the teacher model are obtained via an exponential moving average (EMA) of the student's parameters. Despite the significant advancements achieved by the teacher–student framework in semi-supervised semantic segmentation, recent SOTA solutions still suffer from coupling problems [22]. A clear manifestation of the coupling problem is that, during the initial stages of training, the teacher model can effectively guide the student model, facilitating its learning. However, as training progresses, the weights of the teacher and student models may increasingly converge, leading to a high degree of similarity. This scenario diminishes the teacher model's capacity to provide novel and useful reference signals. Consequently, the training performance of the student model decreases, impacting the overall effectiveness of the method. This phenomenon can be attributed to the fact that the teacher model's parameters are updated using the EMA of the student model's parameters, resulting in a gradual reduction in the

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disparity between their parameters. As training advances, this similarity can escalate, hindering the student model's ability to acquire additional beneficial knowledge from the teacher model. This coupling effect is a critical limitation that needs to be addressed to further advance semi-supervised learning methodologies in the field of semantic segmentation.

To address this challenge, this article introduces a semi-supervised semantic segmentation method, named alternating guidance with cross-view (AGCV). This methodology employs a dual-decoder architecture within the student model, where each decoder shares an identical structural design but maintains independent parameters. This configuration allows the student model to process input images from distinct perspectives, enhancing its ability to capture diverse features and patterns. By promoting self-exploration and learning, this strategy not only improves the model's intrinsic feature extraction capabilities but also fosters a rich and varied representation of knowledge. Simultaneously, we employ the same architecture in the teacher model and implement an alternating supervision mechanism. The teacher model alternates in generating pseudo-labels for the student model, providing a diverse and differential reference signal. This mechanism effectively mitigates the issue of over-similarity between the teacher and student models, thereby enhancing the overall performance and robustness. By integrating these strategic elements, AGCV endeavors to substantially ameliorate the coupling between teacher and student models, thus contributing to a more viable solution to the domain of semi-supervised semantic segmentation for remote sensing images.

Fig. 1 provides a comprehensive comparison of the popular semi-supervised semantic segmentation frameworks, which adopt either implicit or explicit teacher–student approaches [22]. As depicted in Fig. 1(a), approaches such as FixMatch [23], UniMatch [24], and PseudoSeg [25] utilize a single student model, wherein predictions made on weakly augmented images are exploited as pseudo-labels to supervise the predictions on strongly augmented counterparts. The student model processing the weakly augmented input can be conceptualized as a teacher for its counterpart evaluating the strongly augmented input. Subsequently, CPS [14] adopts a dual-student architecture, illustrated in Fig. 1(b), where each of the two models reciprocally serves as a pseudo-label provider for the other, enabling a dynamic interchange of roles between student and teacher. The advent of mean teacher [16] ushered in the era of the explicit teacher–student frameworks, depicted in Fig. 1(c). In this setting, the teacher model generates pseudo-labels to supervise the student model, whose parameters are updated through backpropagation. The teacher's parameters are derived via an EMA of the student's parameters. Several methods, including CutMix-Seg [26], AugSeg [19], and AEL [17], have built upon this foundational teacher–student framework. An extension to mean teacher incorporates an additional auxiliary teacher [27], as shown in Fig. 1(d). Furthermore, PS-MT [21] adopts a dual-teacher strategy, where both teachers collaboratively supervise the student model in an ensemble fashion, as represented in Fig. 1(e). Unlike the above method, our contribution introduces a cross-view alternating guidance mechanism, illustrated in Fig. 1(f). Herein, the student model yields predictions from different views, while the teacher model alternates a supervisory role, thereby mitigating the risk of weight coupling and promoting a more robust learning process.

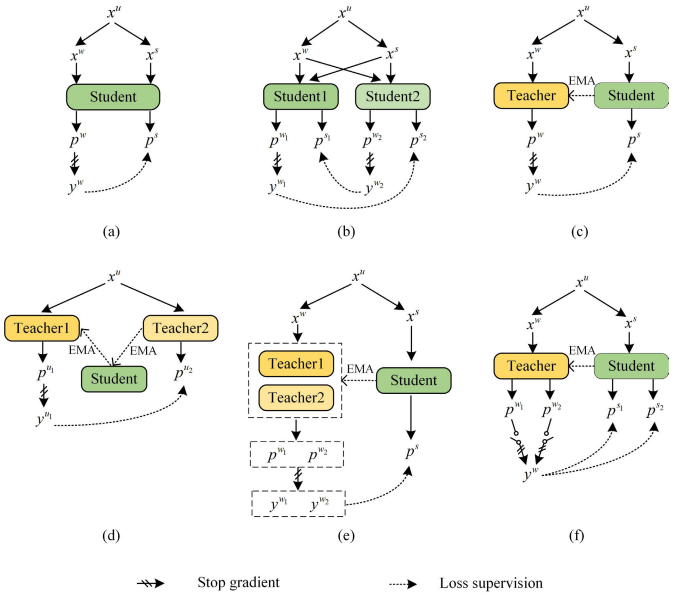


Fig. 1. Comparison among typical semi-supervised semantic supervised frameworks. (a) FixMatch [23], UniMatch [24], and PseudoSeg [25]. (b) CPS [14]. (c) Mean teacher-based structure [16], [17], [19], [26]. (d) GTA [27]. (e) PS-MT [21]. (f) Proposed AGCV. x^u, x^w, and x^s are unlabeled images and unlabeled images with weak and strong perturbations, respectively. p^u, p^w, and p^s represent corresponding predictions for x^u, x^w, and x^s. y^u and y^w are corresponding pseudo-labels for x^u and x^w.

In summary, the main contributions of this article are as follows.

1) A novel method, named AGCV, is proposed for semi-supervised semantic segmentation of remote sensing images. This method effectively alleviates the weight coupling issue between the teacher and student models, thereby significantly enhancing the accuracy of semantic segmentation in a semi-supervised setting for remote sensing imagery.
2) We introduce a dual-decoder architecture within the student model, where each decoder operates independently without parameter sharing. In each iteration, random perturbations are applied to both the feature-level and image-level perspectives. This approach not only encourages the student model to generate diverse feature representations but also enables each decoder to progressively accumulate knowledge related to its assigned view. Thus, the student model acquires batch-specific characteristics, enhancing its ability to capture nuanced patterns and variability in data.
3) The two predictions from the teacher model in the AGCV are alternately exploited as pseudo-labels to guide the training of the student model. This strategy supplies the student model with broad and distinct reference signals, effectively mitigating weight dependency between the teacher and student models and enhancing overall model performance.

The remainder of this article is organized as follows. In Section II, we will discuss related works. Section III will provide details on the proposed AGCV method. Section IV shows the experimental settings, ablation studies, and quantitative and qualitative results. Conclusions are drawn in Section V.

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II. Related Works

This section is devoted to the review of the literature for both semi-supervised learning and semi-supervised semantic segmentation, with a focus on remote sensing.

A. Semi-Supervised Learning

Semi-supervised learning is an active area of research, focused on providing supervision for unlabeled data. The primary paradigms include self-training [11], [28], [29], [30], [31], consistency regularization [32], [33], [34], [35], and their combination [14]. Self-training involves generating pseudo-labels for unlabeled instances, subsequently employing these latter together with ground-truth (GT) labels to retrain the model. Consistency regularization remarks that the model's predictions for the same data should be consistent and stable across various perturbations, including data, features, and network architectures. For instance, the virtual adversarial training (VAT) [33] employs adversarial perturbations to improve model consistency. The dual-student approach [22] further explored the teacher–student paradigm by replacing the conventional teacher role with another student model, refining the learning process. FixMatch [23] leverages the prediction results of weakly augmented images as pseudo-labels to supervise the predictions of strongly augmented images, ensuring pseudo-label quality through a confidence-thresholding strategy. Subsequently, FlexMatch [36] refines this strategy by accounting for the difficulty of learning across different classes. FreeMatch [37] introduced a self-adaptive confidence-thresholding policy. Additionally, DMT [15] exploits two independently initialized models to reweight losses in different regions, achieving yet another innovation in semi-supervised learning.

B. Semi-Supervised Semantic Segmentation

Compared to semi-supervised classification tasks, semantic segmentation demands dense prediction, often requiring more sophisticated methods to achieve accurate performance. Earlier semi-supervised semantic segmentation approaches frequently leveraged generative adversarial networks [38] to generate additional supervisory information for unlabeled data [39], [40], [41]. Self-training is a prevalent method in semi-supervised semantic segmentation, aiming to effectively assign pseudo-labels to unlabeled data and incorporate them into model training as supervisory signals. In this category, ST++ [42] proposes an advanced self-training framework that prioritizes retraining with unlabeled images exhibiting high overall prediction stability. To address the long-tail problem in data distribution, Chen et al. [30] reduced cumulative bias during the pseudo-labeling process from a model design perspective. Recent studies have focused on consistency regularization, for instance, CPS [14] and GCT [13] employ two identical-structured but differently parameterized networks to apply perturbations to the same unlabeled data input. CCT [12] adopts a variety of feature perturbation modalities while imposing constraints to ensure alignment between perturbed and non-perturbed image predictions. Furthermore, the teacher–student paradigm represents a pioneering work in semi-supervised semantic segmentation [16], [18], [20], [21], [27], where only the student model is trained, and the teacher model is updated based on the student model. Specifically, U2PL addressed the issue of unreliable pseudo-labels. GTA [27] introduced an additional teacher model for semi-supervised semantic segmentation. ReCo [43], which is based on the teacher–student framework, enhances learning from unlabeled data through pixel-level contrastive learning. PS-MT [21] exploits dual teacher models to improve prediction accuracy. PCR [20] employs prototype-based predictors to discern intra-class features within unlabeled data. In addition, methods, such as AEL [17], CutMix-Seg [26], AugSeg [19], among others, introduced more advanced data augmentation techniques to further boost model performance.

C. Remote Sensing Semi-Supervised Semantic Segmentation

In recent years, deep learning-driven semi-supervised semantic segmentation has demonstrated potential in remote sensing image analysis, with a series of innovative methods significantly propelling advancements in this domain. BAS4Net [44] integrates a boundary-aware module into a semi-supervised semantic segmentation network, balancing semantic and spatial information. The ClassHyPer [45] approach similarly merges semantic and boundary information, generating semantically coherent mixed images when perturbing unlabeled data, offering a novel perspective on semi-supervised learning. Zheng et al. [46] devised a multiscale convolution module combined with transformer architectures, achieving effective capture of local details and long-range features. The LSST method [47], based on a self-training strategy, selects reliable pseudo-labels via linear sampling for deployment in model training. Building upon consistency regularization, Wang et al. [48] proposed an adaptive pseudo-label updating strategy, enabling more robust utilization of unlabeled data. RanPaste [49] introduced a new form of data perturbation, enhancing data diversity and achieving good results on multiple remote sensing image datasets. Zhang et al. [50] further developed a transformer-based consistency regularization mechanism, ensuring that the model's predictions remain consistent under various random perturbations. The WSCL method [51], through the introduction of sparse bi-view cross-sample image generation technology, not only diversifies the training samples but also fosters the model's understanding and adaptability to complex scenes. ICNet [52], relied upon the foundation of a teacher–student paradigm, proposed a novel method based on iterative contrastive networks. X-ModalNet [53], employing self-adversarial and interactive learning strategies, achieved breakthroughs in cross-modal semantic segmentation tasks for remote sensing images, providing new insights into addressing inter-modal differences and information fusion.

III. Methodology

This section is devoted to the description of the proposed methodology. The preliminaries will be illustrated first. Afterward, the proposed AGCV approach will be presented detailing its main features (i.e., cross-view, random feature perturbation, and alternating guidance). Scalars, vectors, and matrices will be denoted by lowercase or uppercase letters (e.g., x or X), lowercase bold letters (e.g., $\mathbf{x}$), and uppercase bold letters (e.g., $\mathbf{X}$).

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A. Preliminaries

1) Problem Statement: In semi-supervised semantic segmentation, we have a limited labeled dataset, $\mathbf{D^L} = \{[\mathbf{X^L}_i, \mathbf{Y^L}_i]\}_{i=1}^{N^L}$, where $\mathbf{X^L}_i \in \mathbb{R}^{H\times W\times B}$ refers to the ith input image of size $H \times W$ with B spectral bands, $\mathbf{Y^L}_i \in \mathbb{R}^{H\times W\times C}$ is the related GT with C possible classes, $[\cdot,\cdot]$ indicates the matrix concatenation operator along the third dimension, and N^L is the number of images in the $\mathbf{D^L}$ dataset. We also have a large amount of unlabeled data denoted as $\mathbf{D^U} = \{\mathbf{X^U}_i\}_{i=1}^{N^U}$, where N^U is the number of images in the $\mathbf{D^U}$ dataset with $N^U \gg N^L$. Our goal is to use $\mathbf{D^L}$ and $\mathbf{D^U}$ to train a model to generate a dense semantic prediction.

2) Teacher–Student Framework: The teacher–student framework is a widely adopted paradigm in current approaches for semi-supervised semantic segmentation. It consists of a teacher model and a student model. The teacher model generates pseudo-labels by predicting unlabeled images with weak augmentations, which are then used to guide the training of the student model. The student model is trained on unlabeled images with strong augmentations and labeled data. The loss $\mathcal{L}$ for the student model is as follows:

$$\mathcal{L} = \mathcal{L}^L + \mu\mathcal{L}^U \tag{1}$$

$$\mathcal{L}^L = \frac{1}{HWN^L}\sum_{i=1}^{N^L}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C}\left(\mathbf{Y^L}_{i,h,w,c}\log\widehat{\mathbf{Y^L}}_{i,h,w,c}\right) \tag{2}$$

$$\mathcal{L}^U = \frac{1}{HWN^U}\sum_{i=1}^{N^U}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C}\left(\mathbf{Y^U}_{i,h,w,c}\log\widehat{\mathbf{Y^U}}_{i,h,w,c}\right) \tag{3}$$

where $\mathcal{L}^L$ and $\mathcal{L}^U$ are the averaged cross-entropy losses on labeled data and unlabeled data, respectively; $\widehat{\mathbf{Y^L}}_{i,h,w,c}$ and $\widehat{\mathbf{Y^U}}_{i,h,w,c}$ are the student model posterior probabilities for the (h, w)th pixel of the ith image given the cth class on $\mathbf{D^L}$ and $\mathbf{D^U}$, respectively; and $\mathbf{Y^L}_{i,h,w,c}$ and $\mathbf{Y^U}_{i,h,w,c}$ are the related GT and pseudo-label generated by the teacher model, respectively. $\log\cdot$ is the logarithm with base 2, and μ is a weighting parameter. Following [16], [54], we apply a trade-off weight $\mu = e^{-5\times(1-t/t_{\max})^2}$ to control the balance between the supervised loss and the unsupervised loss, where t is the current training iteration and $t_{\max}$ is the maximum number of iterations.

The teacher model and the student model have the same architecture. The final weights of the teacher model, $\boldsymbol{\theta}_t'$, are obtained through an EMA of the student model's weights

$$\boldsymbol{\theta}_t' = \alpha\boldsymbol{\theta}_t + (1-\alpha)\boldsymbol{\theta}_s \tag{4}$$

where $\boldsymbol{\theta}_t$ and $\boldsymbol{\theta}_s$ are the parameters of the teacher and student models, respectively, and α is a hyperparameter in the EMA, with $\alpha \in [0, 1]$.

B. AGCV

The overall architecture of the proposed AGCV approach is depicted in Fig. 2. The student model consists of an encoder and two decoders. $\boldsymbol{\theta}_s = \{\boldsymbol{\theta}_s^g, \boldsymbol{\theta}_s^{h_1}, \boldsymbol{\theta}_s^{h_2}\}$, where $\boldsymbol{\theta}_s^g$, $\boldsymbol{\theta}_s^{h_1}$, and $\boldsymbol{\theta}_s^{h_2}$ indicate the parameters of the encoder and the two decoders of the student model. The teacher model has the same structure as the student model. $\boldsymbol{\theta}_t = \{\boldsymbol{\theta}_t^g, \boldsymbol{\theta}_t^{h_1}, \boldsymbol{\theta}_t^{h_2}\}$, where $\boldsymbol{\theta}_t^g$, $\boldsymbol{\theta}_t^{h_1}$, and $\boldsymbol{\theta}_t^{h_2}$ denote the parameters of the encoder and the two decoders, respectively.

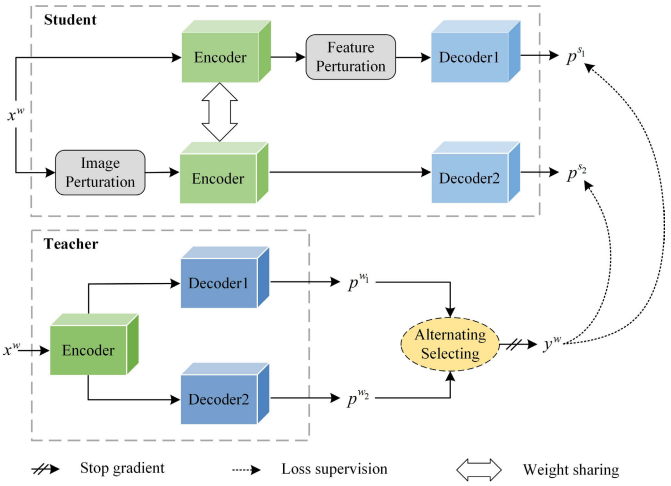


Fig. 2. Illustration of the proposed AGCV approach. In the training process, the student model processes the weakly perturbed unlabeled image x^w using a shared encoder with sharing parameters and two decoders with independent parameters, producing two predictions p^{s_1} and p^{s_2}. Simultaneously, the weakly perturbed unlabeled image x^w is input into the teacher model, which generates two predictions p^{w_1} and p^{w_2}. These two predictions are alternately used as pseudo-label y^w to guide the student model.

1) Cross-View: As illustrated in Fig. 2, the student model incorporates two processing streams: the image perturbation stream and the feature perturbation stream. These streams share a common encoder to ensure consistency in the transformation from the original image to the high-level abstract features. Each stream employs separately initialized decoders with non-shared parameters, allowing the model to parse and reconstruct image information from different perspectives.

For labeled data, after applying weak perturbations, the inputs are fed into both the streams of the student model, and the resulting predictions are supervised by the GT. Thus, we have the following:

$$\mathcal{L}^L = \frac{1}{2}\left(\mathcal{L}_{S_1}^L + \mathcal{L}_{S_2}^L\right) \tag{5}$$

$$\mathcal{L}_{S_1}^L = \frac{1}{HWN^L}\sum_{i=1}^{N^L}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} \times\left(\mathbf{Y^L}_{i,h,w,c}\log\left(h_s^1\left(g_s\left(\mathbf{X^L}\right)\right)_{i,h,w,c}\right)\right) \tag{6}$$

$$\mathcal{L}_{S_2}^L = \frac{1}{HWN^L}\sum_{i=1}^{N^L}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} \times\left(\mathbf{Y^L}_{i,h,w,c}\log\left(h_s^2\left(g_s\left(\mathbf{X^L}\right)\right)_{i,h,w,c}\right)\right) \tag{7}$$

where $g_s(\cdot)$, $h_s^1(\cdot)$, and $h_s^2(\cdot)$ are the encoder and the two decoder functions of the student model, respectively.

For unlabeled data, AGCV showcases cross-view learning, where the two streams of the student model handle perturbed images from both the image-level and the feature-level views. More specifically, the feature perturbation stream, upon acquiring the encoded features of the image, further subjects them to random feature perturbations before passing them to the corresponding decoder to generate the first prediction output. The image perturbation stream receives unlabeled images that have been processed by CutMix. CutMix randomly exchanges image regions yielding a second prediction through the decoder. The definition of the unsupervised loss

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is as follows:

$$\mathcal{L}^U = \frac{1}{2}\left(\mathcal{L}^U_{S_1} + \mathcal{L}^U_{S_2}\right) \tag{8}$$

$$\mathcal{L}^U_{S_1} = \frac{1}{HWN^U}\sum_{i=1}^{N^U}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} \times \left(\mathbf{Y^U}_{i,h,w,c}\log\left(h^1_s\left(g_s\left(\mathbf{X^U}\right)\right)_{i,h,w,c}\right)\right) \tag{9}$$

$$\mathcal{L}^U_{S_2} = \frac{1}{HWN^U}\sum_{i=1}^{N^U}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} \times \left(\mathbf{Y^U}_{i,h,w,c}\log\left(h^2_s\left(g_s\left(\mathbf{X^U}\right)\right)_{i,h,w,c}\right)\right). \tag{10}$$

Through these strategies, AGCV fosters a diversified feature representation learning within the model and enhances its generalization capabilities. It enables the student model to learn cross-view feature models in a more stable way, thereby improving overall segmentation accuracy and robustness.

2) Random Feature Perturbation: During the training phase of the student model, feature perturbation serves as a pivotal component. Inspired by CCT [12], we subject the output features of the encoder, f_1, to six distinct feature perturbations, randomly selecting one at each iteration. The detailed description is as follows.

1) *Dropout:* Random dropout is applied to f_1. Thus, we have $\widetilde{f_1} = \delta(f_1)$, where $\delta(\cdot)$ indicates the dropout function.
2) *Feature Noise:* A Gaussian-distributed random noise is injected into f_1. Initially, a random noise vector N is sampled from a normal distribution, which is then combined with the original features to maintain the feature scale, that is, $\widetilde{f_1} = (N \odot f_1) + f_1$, where $\odot$ is the Hadamard product.
3) *Feature Dropout:* A summation over the channel dimension of f_1 followed by normalization yields a new feature representation, f_1'. Subsequently, a threshold is randomly sampled and the elements of f_1' under this threshold are zeroed out, achieving a selective suppression of features.
4) *Guided Masking:* Considering the importance of contextual relationships in understanding complex scenes, we leverage the teacher model's prediction to identify objects and contextual information. A masking operation is performed on these regions to prevent the model from over-relying on them.
5) *Guided Cutout:* To mitigate the model's dependence on specific regions of targets, we initially exploit the teacher model's prediction to detect the approximate location of each target. Subsequently, a random cropping is conducted on the corresponding areas of f_1. By removing key information, we force the model to extract useful information from the remaining features, supporting its generalization ability.

Through these feature perturbation operations, our student model undergoes a variety of input conditions during training, effectively bolstering its robustness and capability to cope with unknown scenarios. It is worth to be remarked that each perturbation technique is designed to avoid an excessive information loss or the introduction of any unnecessary bias while enhancing model performance.



(a) (b)

Fig. 3. Two instances of remote sensing images from: (a) Yellow River Estuary Wetland dataset and (b) GID.

3) Alternating Guidance: The teacher and student models share the same architecture, as illustrated in Fig. 2, featuring a dual-decoder structure. During training, the teacher model processes weakly perturbed unlabeled images as input, yielding two prediction results that alternately serve as pseudo-labels to guide the two predictions of the student model. This alternating supervision strategy not only ensures that the student model receives a rich set of supervision signals but also dynamically adjusts the source of guidance, effectively alleviating the issue of the weight convergence between the teacher and student models. Notably, when the output from the teacher model's first decoder is used as a pseudo-label to instruct the student model, the parameters of the teacher model's second decoder are frozen, preventing parameter updates, and vice versa. This can be expressed as follows:

$$\boldsymbol{\theta}'_t = \begin{cases} \alpha\left(\boldsymbol{\theta}^g_t, \boldsymbol{\theta}^{h_1}_t\right) + (1-\alpha)\left(\boldsymbol{\theta}^g_s, \boldsymbol{\theta}^{h_1}_s\right), & i \le N^U/2 \\ \alpha\left(\boldsymbol{\theta}^g_t, \boldsymbol{\theta}^{h_2}_t\right) + (1-\alpha)\left(\boldsymbol{\theta}^g_s, \boldsymbol{\theta}^{h_2}_s\right), & i > N^U/2. \end{cases} \tag{11}$$

By providing different predictive outputs to the teacher model, the student model stimulates the teacher model to generate extensive feedback signals for guiding student training, forming a mutually beneficial virtuous learning cycle. This cycle not only facilitates a steady improvement in the performance of the student model but also effectively mitigates the common issue of the weight coupling in traditional teacher–student frameworks. This approach offers novel perspectives and possibilities for the development of semi-supervised semantic segmentation techniques in remote sensing imagery.

IV. Experimental Analysis

In this section, we will provide first a detailed description of both the datasets and the experimental setup (including the adopted evaluation metrics). Subsequently, experiments will be presented considering various remote sensing data and compared the proposed approach with some SOTA methods. Finally, ablation experiments on our AGCV will be provided to the readers.

A. Datasets

To validate the efficacy and generalization ability of the proposed method, this study assesses the performance on the Gaofen image dataset (GID) and the Yellow River Estuary Wetland dataset, as illustrated in Fig. 3.

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1) GID: GID [55] is a large-scale land cover classification dataset derived from observations of China's high-resolution remote sensing satellite, Gaofen-2. This dataset consists of 150 images, each with dimensions of 7200 × 6800 pixels, covering four spectral bands. Within the GID, six major classes of land cover are represented: buildings, farmland, forests, grasslands, water, and other types. Following methodologies from related research [56], we selected a series of image samples ensuring the inclusion of these six categories. By applying a sliding window technique, the original images were cropped into 128 × 128 pixel sub-images, resulting in a total of 20 000 patches. Among these, we randomly selected 80% as the training set, with labeled data constituting 1/4, 1/8, and 1/16 of the total training set, respectively. The remaining 20% of the images were used for the final testing and evaluation of the model's performance.

2) Yellow River Estuary Wetland Dataset: The Yellow River Estuary Wetland dataset is derived from multispectral imaging of the region's wetlands by the Gaofen-1 satellite during different seasons in 2019, as shown in Fig. 3(a). This dataset is categorized into seven ecological types: mudflat, *spartina alterniflora,* salt marsh, tamarix, water, *suaeda salsa,* and reed. The dataset consists of a total of 29 images, each capturing information across four spectral bands: red, green, blue, and near-infrared. We randomly selected six images to form an independent test set to evaluate the model's performance. The remaining 23 images constitute the training set. From the training set, we randomly selected images at ratios of 1/4 (6 images), 1/8 (3 images), and 1/16 (1.5 images) as labeled data. The rest of the training images was used as unlabeled data, participating in the semi-supervised training process of the model.

B. Experimental Setup

1) Implementation Details: We selected the well-established DeepLabV3+ [57] to handle the semantic segmentation task, utilizing the ResNet50 [58] network pre-trained on the ImageNet dataset as the encoder. We conducted the training and testing of the model using the PyTorch framework. To ensure the fairness of the experimental outcomes, identical experimental settings were maintained across all methodologies, retraining them on the two remote sensing datasets employed in this study.

We exploited the SGD optimizer with an initial learning rate set to 0.001 and a weight decay of 0.001. The learning rate for the decoder components was set to be 10 times that of the encoder. The learning rate schedule followed a poly learning rate policy, specifically formulated as $\mathrm{lr} = \mathrm{lr}_0(1 - i/T)^p$, where lr_0 is the initial learning rate, i is the iteration index, T is the total number of iterations, and p is a parameter fixed to 0.9. During training, weak perturbations, such as random flipping and rotation, were applied to the images. For the GID, the training proceeds for 160 epochs, whereas for the Yellow River Estuary Wetland dataset, it spans 120 epochs. The batch size is set to 16.

2) Quality Metrics: Following the standards for quality evaluation [12], [14], we exploited in this work the mean of intersection over union (mIoU) and the overall accuracy (OA) as metrics to assess the performance.

The mIoU metric measures the spatial consistency and overlap between the predicted segmentation masks and the GT. It calculates the average intersection over union for all classes, providing insights into the algorithm's ability to accurately delineate boundaries and capture the extent of land cover classes. Thus, we have the following:

$$\mathrm{mIoU} = \frac{1}{N}\sum \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{12}$$

where TP, FP, TN, and FN represent the true positive, the false positive, the true negative, and the false negative, respectively.

The OA metric measures the percentage of correctly classified pixels across all classes, providing a comprehensive reflection of the algorithm's performance in accurately predicting the correct class labels. It is defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FN} + \mathrm{FP}}. \tag{13}$$

3) Baseline Approaches: To comprehensively evaluate the superiority of the proposed method, we compared it with some SOTA semi-supervised semantic segmentation methods as follows.

1) *U2PL [18]:* U2PL proposed an alternative method for unreliable pseudo-labels by incorporating unreliable predictions as negative samples in the model training process through contrastive learning, thereby improving the utilization of unlabeled data.
2) *ST++ [42]:* ST++ employs a strong data augmentation (SDA) technique together with a selective retraining strategy based on image prediction stability to refine the self-training procedure in semi-supervised semantic segmentation.
3) *GTA [27]:* GTA introduced the use of a gentle teacher model within the teacher–student paradigm to optimize the feature representation transfer, mitigating the negative impact of unreliable pseudo-labels.
4) *Unimatch [24]:* Unimatch leverages a collaborative weak-to-strong consistency strategy by improving training strategies, thus bolstering the model's capacity to effectively learn from unlabeled data.
5) *CCVC [59]:* CCVC relies upon two collaborative learning branches that generate more reliable pseudo-labels by training both the branches to learn from unrelated views and stabilizing the model training process.
6) *WSCL [51]:* WSCL proposes a weak-to-strong consistency learning for semi-supervised semantic segmentation of remote sensing images.
7) *CorrMatch [60]:* CorrMatch uses strategies such as correlation maps to mine more and higher quality regions in unlabeled images, thus achieving more efficient consistency regularization learning.

C. Experimental Results

1) GID: Using the same experimental setting, we compared our method with several SOTA semi-supervised semantic segmentation approaches. We compared the prediction results when the labeled data ratios were 1/4, 1/8, and 1/16. Table I reports the semantic segmentation performance of each model on the GID for different labeled ratios. From the table, it can be seen that AGCV demonstrates significant superiority over the existing (advanced) methods across all labeled data ratios. Remarkably, under the most stringent condition of using only 1/16 of labeled data, AGCV and CCVC are the only methods

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TABLE I
QUANTITATIVE ASSESSMENT FOR THE GID. BEST RESULTS ARE IN BOLDFACE

GID	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
U2PL	74.98	63.52	77.98	67.24	84.21	75.26
ST++	78.90	68.10	81.30	71.75	84.38	75.81
GTA	78.83	68.54	80.12	69.47	82.32	72.67
Unimatch	79.24	68.85	82.96	73.88	84.33	75.91
CCVC	80.02	69.49	82.45	73.41	84.61	75.83
WSCL	79.37	68.88	82.21	72.80	84.71	75.42
Corrmatch	79.86	69.74	83.30	74.20	84.68	76.01
AGCV	**80.52**	**70.32**	**85.03**	**76.95**	**87.37**	**79.83**

TABLE II
QUANTITATIVE ASSESSMENT FOR THE YELLOW RIVER ESTUARY WETLAND DATASET. BEST RESULTS ARE IN BOLDFACE

	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
U2PL	79.22	61.55	80.44	68.75	87.83	74.52
ST++	80.21	58.76	84.52	65.25	88.99	75.72
GTA	81.30	60.47	85.65	67.35	87.15	70.72
Unimatch	85.91	69.34	88.14	73.48	89.01	75.53
CCVC	84.15	66.45	86.70	70.64	87.75	73.36
WSCL	86.06	68.94	87.86	72.53	88.97	74.95
Corrmatch	86.23	69.47	87.30	72.54	88.80	74.86
AGCV	**86.29**	**70.64**	**88.73**	**74.61**	**89.67**	**76.49**

up to an OA of 80%. However, compared to CCVC, AGCV achieves an improvement of 0.5% in mIoU and a nearly 1% increase in OA. This indicates that, even in data-scarce environments, our strategy can more efficiently use available resources to enhance model performance. As the ratio of labeled data increases to 1/8, there is a noticeable improvement in performance across all compared methods. Specifically, CorrMatch and Unimatch exhibit significant enhancements in mIoU. U2PL, ST++, CCVC, and WSCL demonstrate a moderate improvement of approximately 3%, whereas the performance gain for GTA is relatively modest. Notably, AGCV achieves the most substantial improvement and outperforms all other methods. When the ratio of labeled data is further increased to 1/4, the performance of all methods shows a clear enhancement compared to the previous ratio. U2PL, in particular, demonstrates a significant gain. Despite this higher ratio, most methods still fail to surpass the 85% threshold in OA and the 76% threshold in mIoU, except for CorrMatch that performs well under these conditions. In contrast, the AGCV method achieves an mIoU of 79.83% and an OA of 87.37%, highlighting its superior performance in handling higher proportions of labeled data. Overall, the AGCV approach achieved the best performance across all indicators for all proportions. Regardless of whether the labeled data is abundant or scarce, AGCV can effectively leverage the available data to significantly enhance the segmentation performance of the model. This indicates that AGCV can tackle the challenge of data scarcity and can get advantages even when a huge amount of data is available. In summary, although the gains from semi-supervised semantic segmentation methods can be reduced when the amount of labeled data increases, AGCV maintains the highest accuracy across all labeling ratios.

Additionally, Fig. 4 shows that our method has significant advantages over the compared approaches. From the figure, it can be observed that while all the compared methods suffer from some degrees of missed and erroneous extractions, our approach significantly reduces these issues, leading to more accurate segmentation results. These findings corroborate the effectiveness of our approach in leveraging large amounts of unlabeled data for semi-supervised semantic segmentation. Under the same model architecture, our training strategy yields superior performance.

2) Yellow River Estuary Wetland Dataset: Similarly, we compared the performance of all the models on the Yellow River Estuary Wetland dataset using different labeled ratios (i.e., 1/4, 1/8, and 1/16). The results are reported in Table II.

Remote sensing data typically contain richer information with broader coverage and more diverse land cover types than natural images, presenting greater challenges for class identification. The Yellow River Estuary Wetland dataset is more complex regarding land cover types than the GID, particularly due to the boundaries between wetland vegetation that are hard to be distinguished. As shown in Table II, the mIoU index is generally low. U2PL takes pixels in the image that are difficult to determine as negative samples for certain categories. However, there are many uncertain pixels in the Yellow River Estuary Wetland dataset, which increase the difficulty of training the U2PL approach. GTA introduced a mild teacher model to mitigate the impact of unreliable pseudo-labels, but this approach does not work well when dealing with complex datasets as the Yellow River Estuary Wetland one. ST++ emphasizes self-training by gradually selecting reliable pseudo-labels to train the model. However, accurately selecting reliable pseudo-labels on the Yellow River Estuary Wetland dataset is challenging, thus limiting the performance of ST++. WSCL and CCVC achieve a certain level of performance stability by leveraging unlabeled data more deeply. In contrast, CorrMatch obtains satisfactory results on the more complex Yellow River Estuary Wetland dataset through the implementation of label propagation and threshold relaxing. Unimatch, through introducing perturbations of varying degrees to meet consistency constraints, achieves good performance on the Yellow River Estuary Wetland dataset. However, as shown in Table II, although these baseline models perform well, their overall effectiveness remains inferior to that of our proposed model. AGCV encourages the model to focus on the knowledge accumulation in its individual stream by learning images from different perspectives in the student model, thereby enabling the model to better capture the intrinsic features of the image and enhance the generalization ability of the model. In addition, the alternating supervision strategy employed by the teacher model improves the overall performance of the teacher–student framework. These strategies allow AGCV to maintain optimal prediction performance even when dealing with complex and variable land cover environments as in the Yellow River Estuary Wetland dataset. Fig. 5 shows the segmentation results using the Yellow River Estuary Wetland dataset, visually demonstrating the effectiveness of our approach.

D. Ablation Study

To conduct a more in-depth analysis of the effectiveness of our semi-supervised semantic segmentation method and its strategies, we performed ablation studies on two datasets. These experiments covered different ratios of labeled data,

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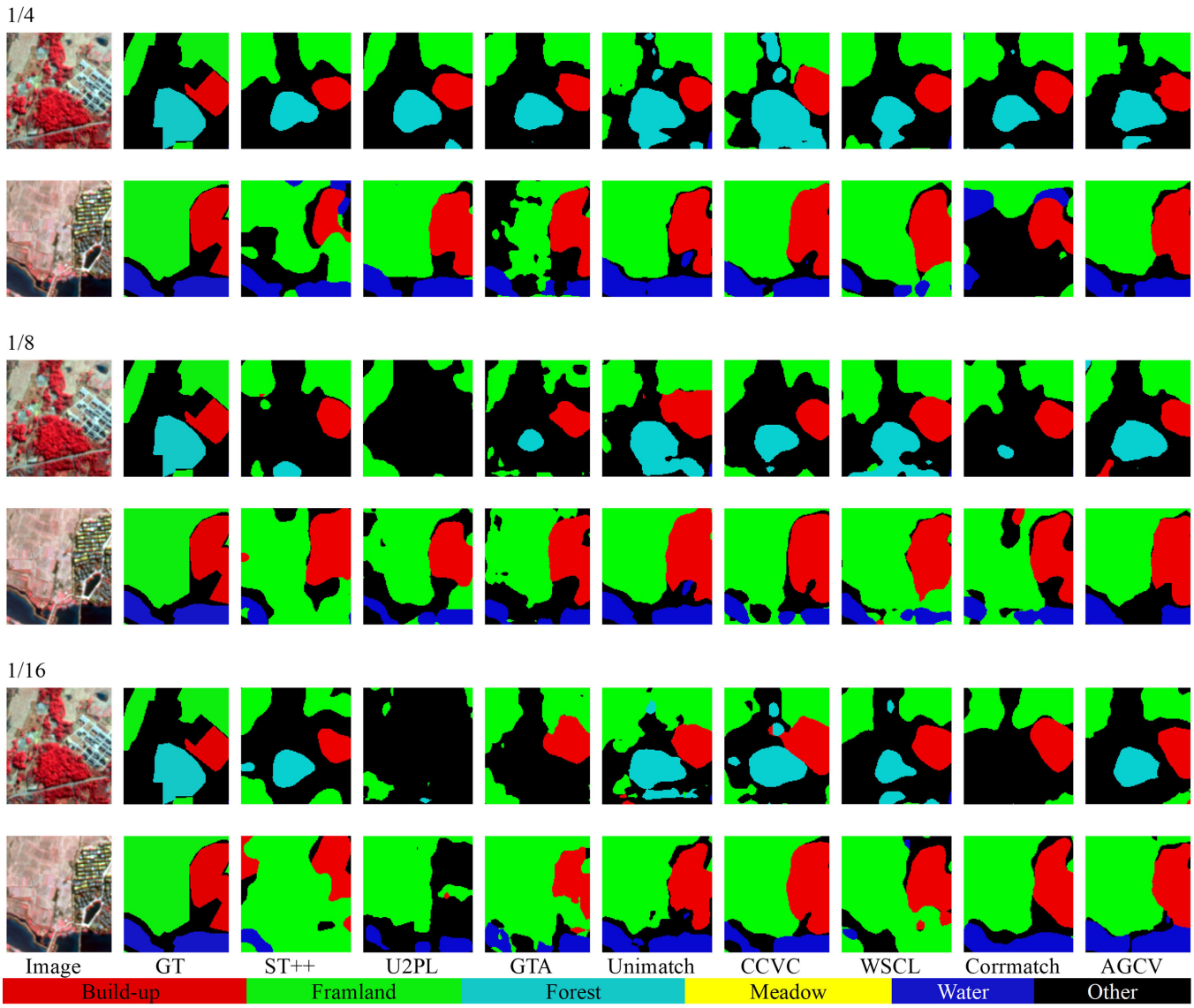


Fig. 4. Visual comparison of the semantic segmentation results for different methods using several ratios of labeled data on the GID. Rows 1 to 2, 3 to 4, and 5 to 6 show the results for the 1/4, 1/8, and 1/16 labeled ratios, respectively.

aiming to comprehensively evaluate the contribution of each strategy to the overall performance.

1) Effectiveness of Dynamic Weight μ: The final loss consists of two parts: supervised loss and unsupervised loss. This article introduces a dynamic weight strategy to balance these two losses. In the early stages of training, the weight of unsupervised loss is lower to allow the model to focus more on labeled data. As training progresses, the model's ability to handle unlabeled data gradually improves, and the weight of unsupervised loss increases, allowing the model to shift its focus to unlabeled data. To verify the effectiveness of this strategy, this article compares it with static weights. The static weight is set to the maximum value of the dynamic weight. The results shown in Fig. 6 indicate that, under different datasets and labeled data ratios, using dynamic weights to balance losses consistently achieves the best segmentation performance compared to static weights.

2) Effectiveness of Alternating Guidance: To validate the effectiveness of the proposed alternating guidance strategy in AGCV, an ablation experiment was conducted where only a single prediction from the teacher model was used as pseudo-labels to guide the training of the student model. The experimental results are reported in Tables III and IV. As can be seen from these tables, compared with the use of a single prediction, the alternating guidance strategy significantly improves the performance of the semi-supervised semantic segmentation. This indicates that alternating the use of predictions from the teacher models as pseudo-labels can more effectively guide the learning process of the student model, thereby enhancing the overall performance of the model.

3) Effectiveness of Cross-View: The student model in AGCV performs a cross-view learning strategy, where one stream executes an image perturbation strategy and the other performs a feature perturbation strategy. To verify the effectiveness of this strategy, the disturbance types used in the two streams of the student model were adjusted. Three strategies were compared: 1) both streams used image perturbation (single image perturbations); 2) both streams use characteristic perturbations (single feature perturbations); and 3) both

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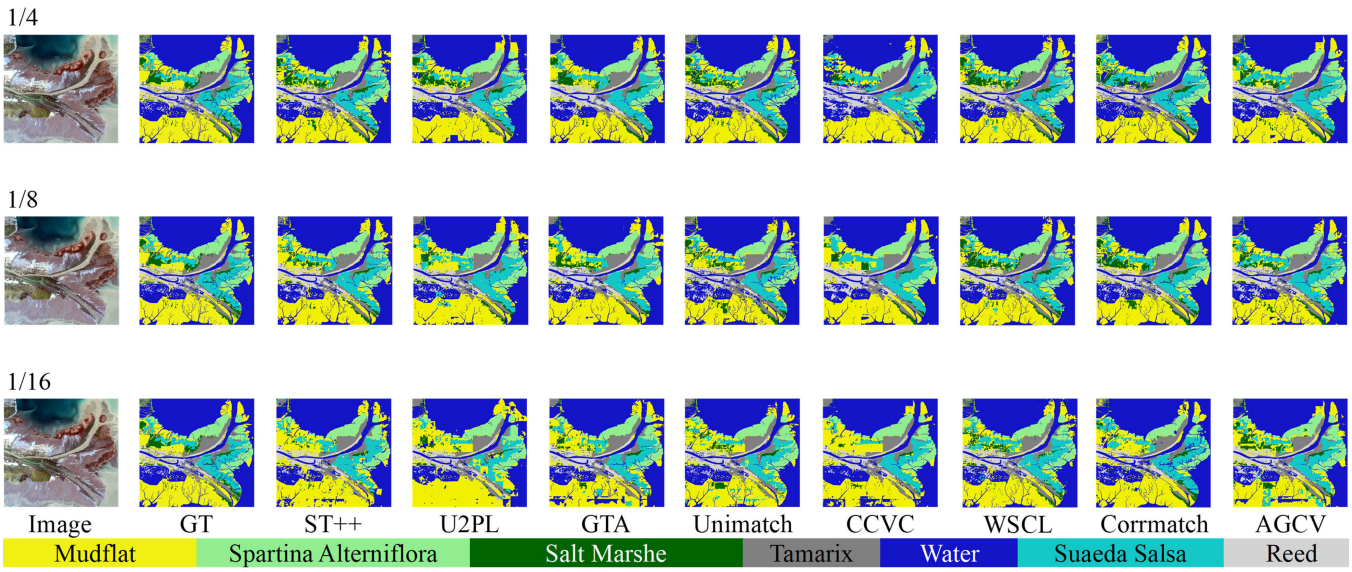


Fig. 5. Visual comparison of the semantic segmentation results for different methods using several ratios of labeled data on the Yellow River Estuary Wetland dataset. Rows 1–3 show the results for the 1/4, 1/8, and 1/16 labeled ratios, respectively.

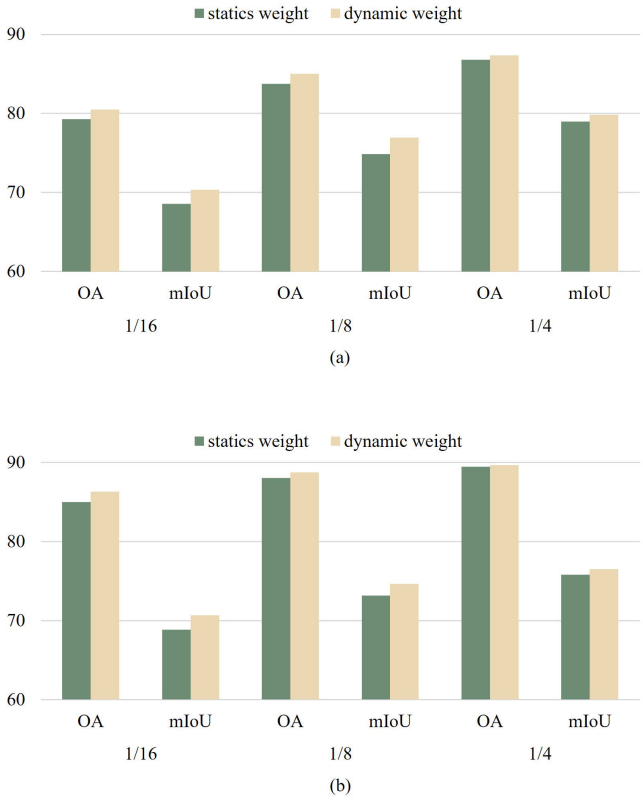


Fig. 6. Ablation study on the efficacy of the dynamic weight μ. (a) and (b) Results for the GID and Yellow River Estuary Wetland datasets, respectively.

streams simultaneously apply image and feature perturbations (dual hybrid perturbations). The results in Tables V and VI indicate that the performance decreases when only a single perturbation strategy (single image perturbation or single feature perturbation) is applied, which confirms the effectiveness of combining the two different perturbation strategies.

TABLE III
ABLATION STUDY ABOUT THE EFFICACY OF THE PROPOSED ALTERNATING GUIDANCE ON THE GID. BEST RESULTS ARE IN BOLDFACE

Method	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
ours	**80.52**	**70.32**	**85.03**	**76.95**	**87.37**	**79.83**
Only decoder 1	79.74	69.14	84.12	75.43	86.11	78.05
Only decoder 2	79.52	69.12	83.74	74.75	86.63	78.88

TABLE IV
ABLATION STUDY ABOUT THE EFFICACY OF THE PROPOSED ALTERNATING GUIDANCE ON THE YELLOW RIVER ESTUARY WETLAND DATASET. BEST RESULTS ARE IN BOLDFACE

Method	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
ours	**86.29**	**70.64**	**88.73**	**74.61**	**89.67**	**76.49**
Only decoder 1	85.64	68.55	87.92	73.61	88.89	75.19
Only decoder 2	85.29	68.82	88.10	72.97	88.92	75.01

In addition, although the segmentation performance obtained by adopting a dual hybrid perturbation strategy is better than that of the single view perturbation, it does not achieve the same performance as when separately applying two types of perturbations. This may be due to the fact that simultaneously applying two perturbations to the input image will greatly increase the learning difficulty of the model, and, therefore, it fails to achieve better segmentation results.

4) Effectiveness of Image Perturbation: The CutMix method [61] is chosen for image perturbation in AGCV. To verify the effectiveness of using CutMix, another common image perturbation technique, ClassMix [62], has been chosen for comparison. We tested two solutions. We replaced CutMix with ClassMix and we randomly selected CutMix and ClassMix for perturbation during training (i.e., random perturbations). The experimental results in Tables VII and VIII demonstrate that except for the random selection strategy using the 1/8 ratio on the Yellow River Estuary Wetland dataset; the performance of using ClassMix or the one of introducing

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TABLE V

ABLATION STUDY ABOUT THE EFFICACY OF THE PROPOSED CROSS-VIEW ON THE GID. BEST RESULTS ARE IN BOLDFACE

Method	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
ours	**80.52**	**70.32**	**85.03**	**76.95**	**87.37**	**79.83**
Dual hybrid perturbations	79.94	69.74	84.19	75.47	86.92	79.24
Single image perturbation	79.89	69.75	84.43	75.83	86.98	79.43
Single feature perturbation	77.30	66.12	82.28	72.62	85.89	77.72

TABLE VI

ABLATION STUDY ABOUT THE EFFICACY OF THE PROPOSED CROSS-VIEW ON THE YELLOW RIVER ESTUARY WETLAND DATASET. BEST RESULTS ARE IN BOLDFACE

Method	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
ours	**86.29**	**70.64**	**88.73**	**74.61**	**89.67**	**76.49**
Dual hybrid perturbations	85.68	69.79	88.28	73.97	88.77	75.04
Single image perturbation	85.33	69.10	88.10	73.64	88.76	74.84
Single feature perturbation	85.49	68.70	88.00	73.28	88.55	74.74

TABLE VII

ABLATION STUDY ABOUT THE EFFICACY OF THE ADOPTED IMAGE PERTURBATION STRATEGY ON THE GID. BEST RESULTS ARE IN BOLDFACE

Method	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
ours	**80.52**	**70.32**	**85.03**	**76.95**	**87.37**	**79.83**
ClassMix	79.54	69.47	83.26	74.17	86.30	78.44
Random perturbations	79.70	69.47	83.39	74.26	87.01	79.37

TABLE VIII

ABLATION STUDY ABOUT THE EFFICACY OF THE ADOPTED IMAGE PERTURBATION STRATEGY ON THE YELLOW RIVER ESTUARY WETLAND DATASET. BEST RESULTS ARE IN BOLDFACE

Method	1/16		1/8		1/4	
	OA	mIoU	OA	mIoU	OA	mIoU
ours	**86.29**	**70.64**	88.73	**74.61**	**89.67**	**76.49**
ClassMix	86.17	69.17	88.60	74.28	89.61	76.28
Random perturbations	86.18	68.93	**88.75**	74.04	89.62	76.03

random perturbations is not as good as using CutMix. This indicates that the perturbation based on CutMix is more suitable for the semi-supervised semantic segmentation method proposed in this article.

5) Effectiveness of Random Feature Perturbation: A strategy based on randomly selecting one of six feature perturbations at each iteration was adopted in AGCV. To verify the goodness of this random selection strategy, it was compared with using a single feature perturbation. The experimental results in Fig. 7 show that only when dropout is applied using a ratio of 1/4, the performance overcomes the one of the proposed strategy on the GID. Instead, on the Yellow River Estuary Wetland dataset, the performance exceeds the current random selection strategy only when the feature dropout is applied using a ratio of 1/8. For all the other perturbation methods, the results were worse than the proposed random selection strategy.

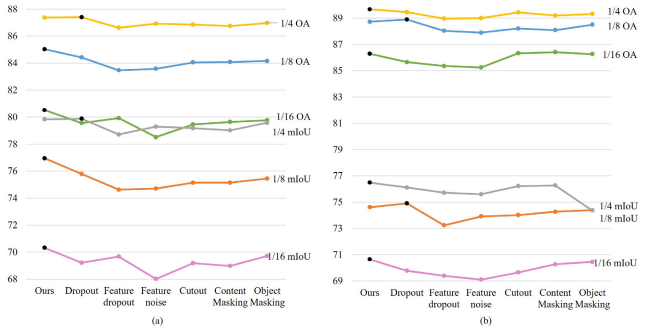


Fig. 7. Ablation study on the efficacy of the random feature perturbation. (a) and (b) Results for the GID and Yellow River Estuary Wetland datasets, respectively. Best results for each metric are indicated with black dots.

V. CONCLUSION

This article proposes a teacher–student method (AGCV) based on cross-view and alternating guidance to enhance the performance of semi-supervised semantic segmentation in remote sensing images. The main goal of this method is to alleviate the weight coupling problem commonly found in teacher–student learning frameworks. In the AGCV method, both the student model and the teacher model adopt a dual-decoder architecture, with both decoders sharing the same architecture but having independent parameter updates. Among them, the student model uses two decoders to process the data obtained through image perturbation and feature perturbation, respectively, enabling the student model to learn the ability to handle different inputs and extract more diverse and unique features from the image. This is beneficial for improving the robustness of student models. Meanwhile, the two decoders of the teacher model take turns generating pseudo-labels to guide the training process of the student model. By providing diverse and differentiated guidance signals for student models, the phenomenon of weight coupling between teacher models and student models can be alleviated. The experimental results show that, compared to the current SOTA semi-supervised semantic segmentation methods, the AGCV method based on the above innovative strategies has achieved the most competitive performance in semantic segmentation.

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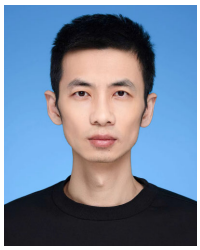
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